Adrian Garcia — Senior AI/ML Engineer

Merced, CA | Open to Remote | Available Immediately

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Portfolio: https://portfolio-adiran-garcia.vercel.app

Senior AI/ML Engineer with 6+ years in applied machine learning, LLMs, and generative AI, and 8+ years in cloud-native backend and full-stack development. I specialize in designing and delivering end-to-end RAG systems, vector search, NLP/transformer pipelines, multi-agent orchestration, and AI-driven analytics platforms with production-grade MLOps, orchestration, and observability. Passionate about healthcare (explainability, PHI/PII protection, auditability) and fintech (fraud detection, payments, compliance), I build scalable APIs, real-time data pipelines, and AI applications with measurable impact on performance, latency, and cost efficiency. My work spans multi-agent AI workflows, conversational AI over global economic datasets, and predictive analytics pipelines, transforming prototypes into reliable, production-ready solutions.

CORE SKILLS

AI/ML & Data Science

- LLMs & RAG: GPT-4/5, LLaMA, Claude, Mistral, LangChain, Pinecone, FAISS, Graph RAG
- NLP & Document Intelligence: BERT/GPT embeddings, OCR/KIE, NER, multi-lingual models, text detection, summarization, Active Learning, Semi-supervised Learning
- Computer Vision: CNNs, YOLOv7, CRNN, image segmentation, autoencoders, GANs
- Model Optimization: LoRA/QLoRA fine-tuning, transfer learning, active/semi-supervised learning, synthetic data generation

Data Engineering & MLOps

- ETL/ELT pipelines, feature engineering, large-scale data curation, Parquet/S3 storage, schema validation
- MLOps: SageMaker, Vertex AI, MLflow, Airflow, Kubeflow, CI/CD, reproducible pipelines, model registry
- Observability & monitoring: Prometheus, Grafana, Datadog, OpenTelemetry, logging (ELK, OpenSearch), evaluation dashboards

Programming & Frameworks

- Python, TypeScript/JavaScript, Go, C++, Rust, Bash/Shell
- PyTorch, TensorFlow, Keras, Scikit-Learn, OpenCV, Hugging Face Transformers, LangChain
- Backend & APIs: FastAPI, Flask, Node.js (Express/NestJS), gRPC, REST, GraphQL
- Cloud & Infrastructure: AWS (ECS, EKS, Lambda, Glue, Bedrock), GCP, Azure, Kubernetes, Docker, Terraform, CloudFormation

Databases &Storage

- SQL/NoSQL: PostgreSQL, Redis, Cassandra, Oracle, Neo4j
- Vector DBs: Pinecone, FAISS, Chroma DB
- Data lakes & blob storage: Azure Blob, Azure Data Lake, S3

Frontend & Data Visualization

- React, Next.js, Chart.js, D3.js
- API-driven UIs, accessibility compliance (WCAG), performance-optimized dashboards

Collaboration & Security

- Agile/Scrum, Jira/Confluence, code review, team mentorship
- Security & compliance: OAuth2, JWT, OIDC, IAM, secure cloud deployments

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PROFESSIONAL EXPERIENCE

Harvest LLC ▪ Wilmington, DE —*Backend & AI Engineer (Contract)*
Remote | Apr 2025 – Oct 2025

RAG/Conversational AI

- Developed a LangChain + Pinecone RAG stack over WDI, Eurostat, and IMF datasets, combining hybrid retrieval (BM25 + dense embeddings), reranking, and source citations, delivering accurate, verifiable answers in real time.
- Implemented multilingual QA pipelines using OpenAI embeddings and GPT APIs, including guardrails for prompt-injection mitigation, citation validation, and "fail-closed" responses, reducing incorrect or unsafe responses by 35%.
- Added voice-based Q&A with STT → reasoning → TTS pipelines, integrated into web and telephony interfaces, cutting time-to-answer for phone users by 50%.

Data Pipelines & Quality

- Built robust ETL/ELT pipelines for gigabyte-scale sources with incremental pulls, deduplication, and schema-drift detection, reducing ingestion failures and backfill efforts by 40%.
- Standardized storage with S3/Parquet + Postgres metadata, implementing lineage and reproducible snapshots, ensuring full auditability and regulatory compliance.

MLOps & Reliability

- Designed evaluation dashboards tracking faithfulness, relevance, groundedness, latency, and cost per query, enabling data-driven optimization of prompts and retrieval strategies.
- Managed MLflow model registry with blue-green and canary deployments, reducing production incidents and ensuring continuous availability of ML services.
- Implemented CI/CD pipelines (GitHub Actions) with unit/contract tests and image scanning, cutting deployment errors by 60%.

Cloud Backend & Platform Engineering

- Developed FastAPI microservices with idempotent endpoints, JWT/OAuth2 authentication, rate-limiting, and schema validation; exposed OpenAPI contracts for partner integrations.
- Containerized and deployed services on AWS ECS/EKS with auto-scaling and zero-downtime rollouts; integrated OpenTelemetry for observability, improving p95 latency and cost visibility by 30%.

Citadel LLC ▪ Chicago, IL, US— ***Software Engineer (AI/ML, Part-Time)***

Remote | Dec 2023 – Mar 2025

Automation & Workflow Engineering

- Built end-to-end automation pipelines for financial and document workflows using Selenium, RPA, and cloud-native orchestration, reducing manual effort and improving deployment reliability within CI/CD.
- Developed real-time monitoring dashboards with cloud logging and alerting, increasing visibility and operational uptime.
- Designed scalable automation systems using Kubernetes, Docker, and Terraform, enabling high-volume tasks such as payment reconciliation, compliance checks, and document validation.
- Improved automation reliability with containerized Selenium test environments integrated into Jenkins and GitLab CI/CD pipelines.
- Automated document ingestion and preprocessing pipelines using Azure AI and cloud functions, embedding unstructured files into vector databases for downstream retrieval.

AI, NLP & Document Intelligence

- Built intelligent document-parsing services powered by GPT-4o-mini, BERT-GCN KIE models, and NER pipelines, deployed as Kubernetes microservices.
- Integrated AI/ML models with RPA bots to power cognitive automation, improving decision accuracy in document-heavy workflows.
- Deployed Llama2 and other LLM-driven content generation systems in production using LangChain and REST APIs.
- Developed and optimized CNN-based OCR systems with TensorRT and Docker, reducing inference latency by 41%.

Data Engineering & MLOps

- Maintained production data pipelines across PostgreSQL, Oracle, Redis, and Cassandra, ensuring consistent and reliable data flow.
- Designed and automated MLOps workflows using Terraform, Jenkins, MLflow, GitOps, and Vertex AI, improving model traceability and lifecycle management.
- Built synthetic data generation frameworks with PyTorch, OpenCV, and Scipy to enhance KIE model training.
- Accelerated large-scale training and fine-tuning using SageMaker and CUDA-based workflows.
- Managed Airflow, Kubeflow, and Jenkins pipelines for large-scale web scraping and enterprise document indexing tasks.

Retrieval & Search Systems (RAG)

- Engineered cloud-native RAG pipelines using vector databases and Kubernetes, improving contextual search quality and latency.
- Developed AI-driven document search systems using Azure Blob Storage, modern embedding models, and performant REST APIs.
- Implemented Graph RAG solutions with Neo4j on Kubernetes to support relationship-aware analytics and enhanced retrieval.

Security, Observability & Collaboration

- Managed Istio service mesh to strengthen cluster security, traffic routing, and service observability.
- Designed collaborative ML environments using Databricks and JupyterHub for distributed experimentation and rapid iteration.
- Maintained and optimized CI/CD pipelines with GitHub, Jenkins, and Datadog for automated deployment and performance monitoring.

High-Performance Engineering

- Delivered sub-second ML inference in production using GPU-accelerated models deployed on Kubernetes clusters.
- Designed secure, cloud-native ML pipelines with AWS Glue, SageMaker, and vector stores, achieving up to 45% efficiency gains and 30% cost reduction through Terraform and Docker automation.

Convoy Inc ▪ Seattle, WA — *Senior AI & GenAI Engineer (Full-Time)*
Remote | Oct 2022 – Nov 2024

RAG Pipeline Architecture & Document Processing

- Architected an end-to-end document ingestion pipeline using Python and Apache Airflow to process legal PDFs and HTML content, applying spaCy-based text segmentation and generating semantic embeddings with Sentence Transformers (all-MiniLM-L6-v2), then indexing them in FAISS for high-throughput vector search.
- Engineered dense retrieval workflows by encoding user queries using the same embedding model, executing top-k similarity search in FAISS, and applying cross-encoder reranking to maximize context precision and retrieval quality.
- Designed advanced prompt-augmentation strategies in LangChain, dynamically constructing optimized LLM prompts by blending retrieved context with user queries while enforcing token-budget constraint.

LLM Integration & Context-Grounded Generation

- Integrated HuggingFace Transformer models (GPT-3, BERT, Electra) into retrieval pipelines to deliver context-grounded, legally compliant responses by combining parametric model knowledge with retrieved document insights.
- Built and deployed scalable RAG-driven Q&A microservices using FastAPI, encapsulating retrieval, prompt construction, and generative reasoning into low-latency, production-ready APIs.
- Implemented automated knowledge base refresh pipelines via AWS Lambda and S3 event triggers, allowing continuous ingestion, re-embedding, and FAISS index updates without service interruption.

Scalable AI Infrastructure & Deployment

- Designed and optimized GPU-accelerated inference environments using Docker, Kubernetes, and TensorRT, enabling high-throughput, low-latency execution for LLM and embedding workloads.
- Implemented distributed processing pipelines for large-scale embedding generation and vector indexing using Airflow, AWS Batch, and parallelized Python workers, significantly reducing end-to-end processing time.

- Built secure, modular infrastructure-as-code (IaC) deployments using Terraform and AWS services (ECS, EKS, Lambda), ensuring reproducible, fault-tolerant AI environments that support continuous model updates.

Evaluation, Governance & Security

- Developed evaluation frameworks to track retrieval quality and LLM accuracy using recall, BLEU, ROUGE, and custom scoring metrics, with human-in-the-loop review processes ensuring correctness in legal workflows.
- Secured RAG infrastructure—vector databases, embedding pipelines, and inference endpoints—using OAuth2 authentication, encrypted storage, and strict access policies to protect sensitive legal documents.
- Established observability and monitoring for embedding drift, retrieval effectiveness, and generation consistency, enabling proactive model improvements and governance across the RAG ecosystem.

Zillow ▪ Settle, WA — ***Senior Software Engineer (Data Science, AI, Full-Time)***
Remote | Jun 2019 – Oct 2022
ML Systems Engineering & Applied Predictive Modeling

- Led the development of data-driven pricing and resale optimization systems for Zillow's iBuyer business, delivering measurable improvements in fee-setting accuracy and operational efficiency.
- Designed, implemented, and deployed an ensemble of production-ready ML models that automated customer home-pricing workflows at scale, significantly accelerating offer generation.
- Built a human-in-the-loop recommender system to support analysts in constructing Comparative Market Analyses (CMAs), improving decision quality and reducing manual workload.

Deep Learning Solutions for Vision & Text Processing

- Engineered advanced deep learning architectures—including CNNs for document image classification/segmentation and RNNs (LSTM, GRU) for sequential text modeling—enabling robust processing of scanned documents and system logs.
- Improved multilingual policy analysis workflows by implementing encoder–decoder models with attention, enhancing translation and summarization accuracy across five languages.
- Developed autoencoders and variational autoencoders (VAEs) for dimensionality reduction and anomaly detection, enabling early identification of data inconsistencies across millions of records.

Model Optimization, Transfer Learning & Synthetic Data Engineering

- Applied transfer learning with pre-trained CNNs (VGG, ResNet, Inception, DenseNet), improving accuracy and reducing training time for document-image tasks.
- Boosted model robustness by designing and evaluating ensemble techniques—including boosting, bagging, stacking, and random forests—across diverse document and text data types.
- Implemented GAN-based synthetic data generation pipelines to enrich rare document classes, improving downstream model generalization.

Scalable ML Infrastructure, Training & Evaluation

- Built scalable training and inference pipelines using TensorFlow, PyTorch, and distributed GPU clusters, reducing experimentation cycles and enabling rapid model deployment.
- Performed comprehensive hyperparameter tuning and neural architecture search (grid, random, Bayesian optimization) to maximize predictive performance and efficiency.

- Established rigorous evaluation frameworks using cross-validation, ROC-AUC, F1-score, and precision–recall metrics to ensure production-grade model reliability.

Data Collaboration, Feature Engineering & Cross-Functional Impact

- Collaborated with product managers, analysts, and domain experts to design high-quality feature engineering pipelines that improved interpretability and accuracy of predictive models.
- Translated business requirements into scalable ML system designs, ensuring reliability, maintainability, and integration with downstream workflows.
- Provided technical leadership across ML lifecycle stages—model design, code review, deployment, monitoring—and mentored junior engineers on best practices in applied ML and data engineering.

Twills ▪ NY, NY, US— ***Full Stack Engineer (On-Site, Part-Time, Full-Time)***
Remote | Nov 2017 – Apr 2019
Backend Engineering (Node.js & Python)

- Re-architected and scaled the company's auto-parts e-commerce backend using Node.js/Express and Python-based services, delivering reliable ordering, account, and product-catalog APIs supporting 1,000+ daily users.
- Implemented enterprise-grade security: JWT/OAuth2 authentication, rate limiting, idempotent webhooks, RBAC, audit logging, and strict request validation to safeguard sensitive data and transactions.
- Developed reusable service integrations in Node.js/Python for payments, shipping carriers, and inventory systems, using standardized OpenAPI-defined API contracts to ensure consistency across partners.

API Development, Data Systems & Performance Optimization

- Designed and maintained high-availability B2B and internal APIs enabling real-time data exchange between back-office systems, customer-facing dashboards, and external vendor feeds—with automated monitoring and SLA alerting.
- Optimized MySQL (InnoDB) through advanced indexing, query plan tuning, connection pooling, and read/write segregation, significantly reducing p95 latency and overall CPU utilization.
- Implemented background workers, Celery/queue-based pipelines, and asynchronous job processors for imports, reconciliation, and email workflows—improving responsiveness of user-facing features.

Cloud, DevOps & Reliability Engineering

- Led migration from on-prem infrastructure to a containerized, cloud-ready environment using Docker, VMware, and CI/CD pipelines, achieving predictable and low-downtime deployments.
- Added centralized logging, metrics, health checks, and incident alerting, greatly improving system observability and proactive issue detection.
- Established automated backup/restore processes, disaster recovery playbooks, test automation, and environment-parity standards to strengthen platform stability.

Frontend Engineering (React & Next.js)

- Modernized internal admin dashboards and catalog management tools using React and Next.js, delivering improved workflows for search, pricing, and inventory operations.
- Built responsive, API-driven UIs aligned with WCAG accessibility guidelines, ensuring maintainability, performance, and a strong user experience.

- Collaborated closely with designers and backend teams to maintain seamless data flow, intuitive interactions, and consistent end-to-end performance.

EDUCATION

- **UC Irvine —** PhD, Material Science **with Concentration in Computational Science / Data Science** · 2014-2019

  Research in simulation/data analysis; heavy use of **Python**, **C/C++**, **NumPy/Pandas**, **ML/Stats**, **HPC** (MPI/SLURM)

- **UC Irvine —** MS, MSE **(Scientific Computing & Analytics)** · 2012-2014

  Relevant CS topics: **scientific computing**, **numerical methods**, **data processing**, **visualization**

- **UC Merced —** BS, ME **(Programming & Modeling focus)** · 2008-2012

  Emphasis on **computational modeling** and **programming** (MATLAB/**Python**/**C++**)

PERSONANL PROJECTS

- **DataHaven.tech — AI Data Visualization Platform (MVP).**
  Designed and developed an intelligent analytics platform enabling real-time, data-driven Q&A across large-scale economic datasets. The system is powered by a Go-based microservices backend, FastAPI-driven AI services, and a React visualization layer using D3.js and Chart.js for interactive insights and advanced visual analytics. Integrated and normalized high-volume data pipelines from major global economic sources—including World Development Indicators (WDI), International Monetary Fund (IMF) datasets, and Eurostat—ensuring consistent schemas, efficient indexing, and fast query performance. Delivered a fully operational frontend while migrating backend services, vector search pipelines (Pinecone), and LLM-based reasoning components (OpenAI) toward a scalable, production-ready architecture.

- **Company Data Analytics — End-to-End Data Analytics & Predictive Modeling Pipeline.**
  Built a complete analytics pipeline to scrape company data from Glassdoor and corporate websites, perform data cleaning and feature engineering, and generate semantic embeddings using a local Ollama LLM. Orchestrated all workflows with Python-based Airflow DAGs, incorporating task dependencies, automated retries, alerting, and self-healing reprocessing for failed embedding tasks. Developed predictive models (Logistic Regression, Gradient-Boosted Trees) to evaluate company stability, investment risk, and growth potential, and automated reporting via Jupyter/Quarto notebooks to produce clear insights, visualizations, and model explanations.

- **Multi-Orchestrator —Multi Orchestrator pipeline.**
  Developed a FastAPI + asyncio backend to coordinate multi-agent pipelines defined as DAGs, enabling parallel execution, conditional task routing, and fault-tolerant recovery. Implemented a task router, async DAG orchestrator with retries/backoff, and a cross-output verifier for consistency checks. Built an evaluation harness with metrics tracking (precision/recall, latency) and integrated structured logging, SSE-based monitoring, and CI-driven regression tests. Packaged and deployed using Docker with full unit and integration test coverage.

LANGUAGE

- English: Native
- Spanish: Conversational